Algorithm 1 General Approach for Graph G

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lb = find lower bound (G)
preprocessing (G, lb)
ub = find upper bound (G)
for k = ub − 1, ..., lb do
    Initialize SAT (G, k)
    sat = Solve SAT
    while sat = true and ∃bicolored cycle do
        Update SAT (G, k, cycle)
        sat = Solve SAT
    end while
    if sat = false then
        return k − 1
    end if
end for
return lb
```

We have several ways of finding cycles. Though all rely on a simple BFS.

1 Assignment model

Have $n \cdot k$ variables. With $x_{v,i} = \begin{cases} 1 & \text{if } v \text{ has the color } i \\ 0 & \text{else} \end{cases}$. Then we have the following conditions for standard coloring:

$$\sum_{i=0}^{k} x_{v,i} = 1 \qquad \forall v \in V$$
$$x_{v,i} + x_{w,i} \leq 1 \forall \{v, w\} \in E, \forall i \in \{1, ..., k\}$$

As logic formula this corresponds to the conjunction of the following clauses

$$\bigvee_{i=0}^{k} x_{v,i} \qquad \forall v \in V$$
$$\neg x_{v,i} \vee \neg x_{v,j} \quad \forall v \in V, \forall i < j \in \{1, ..., k\}$$
$$\neg x_{v,i} \vee \neg x_{w,i} \forall \{v, w\} \in E, \forall i \in \{1, ..., k\}$$

To forbid a cycle $C = (c_1, ...c_l) \subseteq V$, we then add the clauses

$$\bigvee_{s=1,3,5,..l-1} \neg x_{c_s,i} \vee \bigvee_{s=2,4,6,..l} \neg x_{c_s,j} \forall i \neq j \in \{1, ..., k\}$$

2 Partial-ordering model

Have $n \cdot k$ variables. We assume an ordering on the colors (by naming them in $\{1, ..., k\}$) and then $x_{v,i} = \begin{cases} 1 & \text{if } v \text{ has a color bigger than } i \\ 0 & \text{if } v \text{ has a color smaller or equal to } i \end{cases}$. So every vertex will be assigned the color with $x_{v,i-1} - x_{v,i} = 1$. Then we have the following conditions for standard coloring:

$$\begin{aligned}
&x_{v,k} = 0 && \forall v \in V \\
&x_{v,i} - x_{v,i+1} \geq 0 && \forall v \in V, \forall i \in \{1, ..., k-1\} \\
&x_{v,1} + x_{w,1} \geq 1 && \forall \{v, w\} \in E \\
&x_{v,i-1} - x_{v,i} + x_{w,i-1} - x_i \leq 1 \forall \{v, w\} \in E, \forall i \in \{2, ..., k\}
\end{aligned}$$

As logic formula this corresponds to the conjunction of the following clauses

$$\begin{aligned}
&\neg x_{v,k} && \forall v \in V \\
&x_{v,i} \vee \neg x_{v,i+1} && \forall v \in V, \forall i \in \{1, ..., k-1\} \\
&x_{v,1} \vee x_{w,1} && \forall \{v, w\} \in E \\
&\neg x_{v,i-1} \vee x_{v,i} \vee \neg x_{w,i-1} \vee x_i \forall \{v, w\} \in E, \forall i \in \{2, ..., k\}
\end{aligned}$$

To forbid a cycle $C = (c_1, ...c_l) \subseteq V$, we then add the clauses

$$\bigvee_{s=1,3,5,..l-1} \neg x_{c_s,i-1} \vee x_{c_s,i} \vee \bigvee_{s=2,4,6,..l} \neg x_{c_s,j-1} \vee x_{c_s,j} \forall i \neq j \in \{1, ..., k\}$$

though, if one color w.l.o.g i is 1, there is no $x_{c_s,i-1}$ so leave that part out and just require $x_{c_s,i}$.